Consider the following optimization problem:

$$\min f(x_1, x_2, x_3, x_4) = x_1^2 + \sin(x_3 - x_2) + x_4 + 1$$

subject to

$$x_1 + x_2 + x_3 = 1,$$
$$x_1 - x_2 + x_4 = 1.$$

Use two different ways to eliminate the constraints. For each case, identify the basic and non basic variables.